

Personalized Networking Assistant

Project Description

The **Personalized Networking Assistant** is an AI-powered web application designed to help users build meaningful professional and social connections by generating intelligent, context-aware conversation starters. The application analyzes event descriptions using **DistilBERT** to identify important themes and leverages **GPT-2** to generate personalized conversation suggestions based on the event context and user interests.

To improve the reliability of conversations, the system integrates the **Wikipedia API** for real-time fact verification. Users can also maintain a history of previous conversations and provide feedback, enabling continuous improvement of future recommendations. The application is developed using **FastAPI, Streamlit, Python, HTML, and CSS**, providing a responsive and user-friendly interface for seamless interaction.

Scenarios

Scenario 1

A user attending a technology conference enters the event description into the application. The system analyzes the content and generates personalized conversation starters related to Artificial Intelligence, Cloud Computing, and emerging technologies, helping the user confidently begin meaningful discussions.

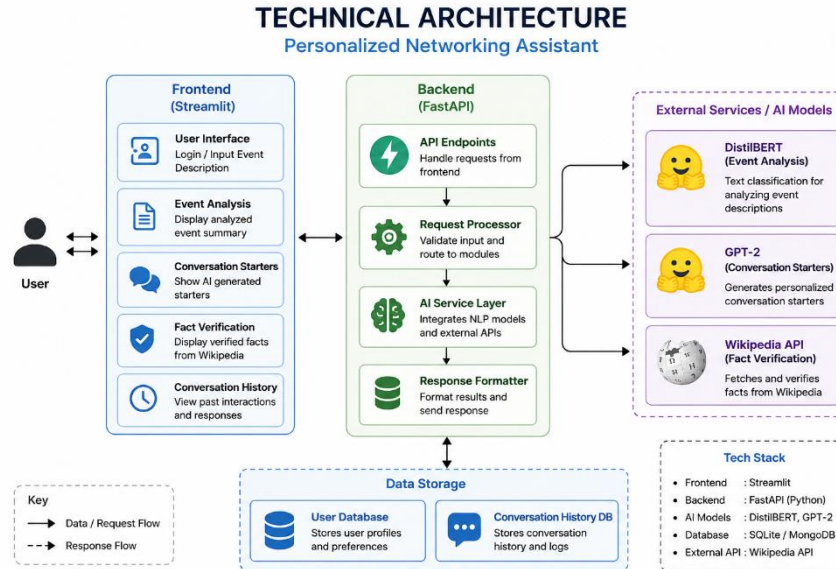
Scenario 2

A professional preparing for a networking event receives AI-generated conversation suggestions tailored to their interests and the event agenda. The application also verifies important facts using the Wikipedia API, ensuring the user can engage in informed and accurate conversations.

Scenario 3

After completing a networking session, the user saves the conversation history and submits feedback. Based on previous interactions, the system improves future recommendations and generates more personalized networking suggestions.

Technical Architecture:



Description

The **Personalized Networking Assistant** follows a modular AI-powered architecture. The frontend, built using **Streamlit**, collects user inputs such as event descriptions and networking preferences. The **FastAPI** backend processes these requests and communicates with **DistilBERT**, **GPT-2**, and the **Wikipedia API** to analyze event content, generate personalized conversation starters, recommend networking topics, and verify factual information. The application stores conversation history and user feedback to improve personalization while ensuring a responsive and scalable user experience.

Pre-requisites

- Python Programming Knowledge
- FastAPI Framework
- Streamlit Framework
- DistilBERT Model
- GPT-2 Model
- Wikipedia API
- HTML & CSS
- Git & GitHub
- VS Code Development Environment
- Render Deployment Knowledge

Project Workflow

Milestone 1 – Requirement Analysis and Planning

- Identify networking challenges and define project objectives.
- Analyze functional and non-functional requirements.
- Select DistilBERT and GPT-2 as AI models.
- Design system architecture and technology stack.

Milestone 2 – Backend Development

- Develop FastAPI backend.
- Integrate DistilBERT for event analysis.
- Integrate GPT-2 for conversation generation.
- Implement Wikipedia API for fact verification.

Milestone 3 – Core Feature Development

- Develop Event Analysis module.
- Implement Personalized Conversation Generator.
- Develop Recommendation Engine.
- Implement Conversation History and Feedback.

Milestone 4 – Frontend Development

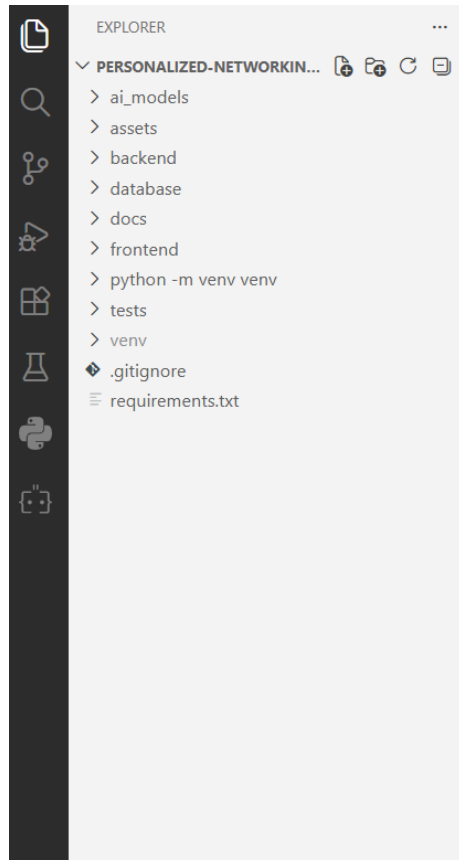
- Develop Streamlit-based user interface.
- Create responsive pages.
- Integrate frontend with backend APIs.

Milestone 5 – Testing & Deployment

- Perform functional and performance testing.
- Fix bugs and optimize performance.
- Deploy the application on Render.
- Prepare documentation and GitHub repository.

Milestone 1: Requirement Analysis & Planning

The first phase focused on understanding the challenges faced by users during networking events. The project requirements were analyzed, and the objectives were defined to build an AI-powered assistant capable of generating personalized conversation starters, recommending discussion topics, and verifying facts. The technology stack, architecture, and development plan were finalized.



Milestone 2: Backend Development

The backend was developed using FastAPI to process user requests and manage application logic. APIs were implemented to analyze event descriptions, generate AI-powered conversation starters, and interact with the database. Proper error handling and modular coding practices were followed.

EXPLORER

PERSONALIZED-NETWORKIN...

ai_models

assets

backend

__pycache__

__init__.py

app.py

config.py

models.py

routes.py

utils.py

database

docs

frontend

python -m venv venv

tests

venv

.gitignore

requirements.txt

app.py

```

1 from fastapi import FastAPI
2 from backend.routes import router
3
4 app = FastAPI(
5     title="Personalized Networking Assistant",
6     version="1.0.0"
7 )
8
9 app.include_router(router)
10
11
12 @app.get("/")
13 def home():
14     return {
15         "message": "Welcome to Personalized Networking Assistant"
16     }
17
18
19 @app.get("/health")
20 def health():
21     return {
22         "status": "Running"
23     }

```

Personalized Networking Assistant 1.0.0 OAS 3.1

/openapi.json

default

GET	/hello	Hello	▼
POST	/user	Create User	▼
GET	/users	Get Users	▼
POST	/analyze	Analyze Event	▼
POST	/generate	Generate	▼
GET	/fact-check/{topic}	Verify	▼
GET	/history	History	▼
GET	/feedback	Get Feedback	▼
POST	/feedback	Feedback	▼
GET	/	Home	▼

Milestone 3: AI Feature Development

The AI modules were integrated using DistilBERT, GPT-2, and the Wikipedia API. DistilBERT extracts important themes from event descriptions, GPT-2 generates personalized conversation starters, and Wikipedia API verifies information. These features improve networking quality and user confidence.

Deploy ⋮

Submit Feedback

✓ Feedback Saved Successfully!

Fact Checker ↗

Enter Topic to Verify

Artificial Intelligence

Check

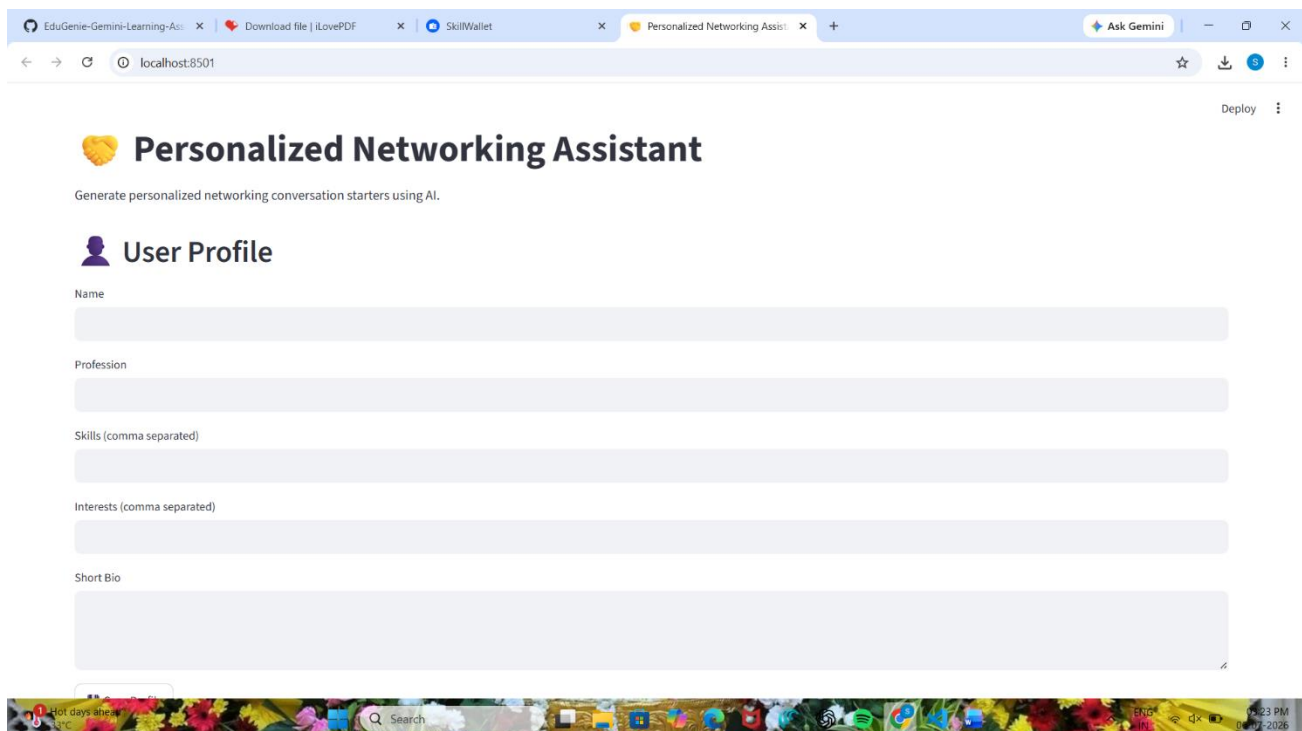
Fact Check Result

Verified

Artificial intelligence (AI) is the capability of computational systems to perform tasks typically associated with human intelligence, such as learning, reasoning, problem-solving, perception, and decision-making. It is a field of research in engineering, mathematics and computer science that develops and studies methods and software that enable machines to perceive their environment and use learning and intelligence to take actions that maximize their chances of achieving defined goals. High-profile applications of AI include advanced web search engines, chatbots, virtual assistants, autonomous vehicles, and play and analysis in strategy games (e.g., chess and Go).

Milestone 4: Frontend Development

The frontend was developed using Streamlit to provide a simple and responsive interface. Users can enter event details, receive AI-generated conversation suggestions, verify facts, and view conversation history through an interactive dashboard.



 Event Details

Event Title

AI Innovation Summit 2026

Event Description

The AI Innovation Summit 2026 brings together AI researchers, software engineers, machine learning experts, cloud architects, data scientists, and technology enthusiasts. The event includes keynote speeches, panel discussions, technical workshops, networking sessions, startup showcases, and career guidance. Participants will explore Artificial Intelligence, Machine Learning, Generative AI, Cloud Computing, Data Science, and automation while connecting with industry professionals.

Analyze Theme

Generate Conversation

Conversation History

Feedback History

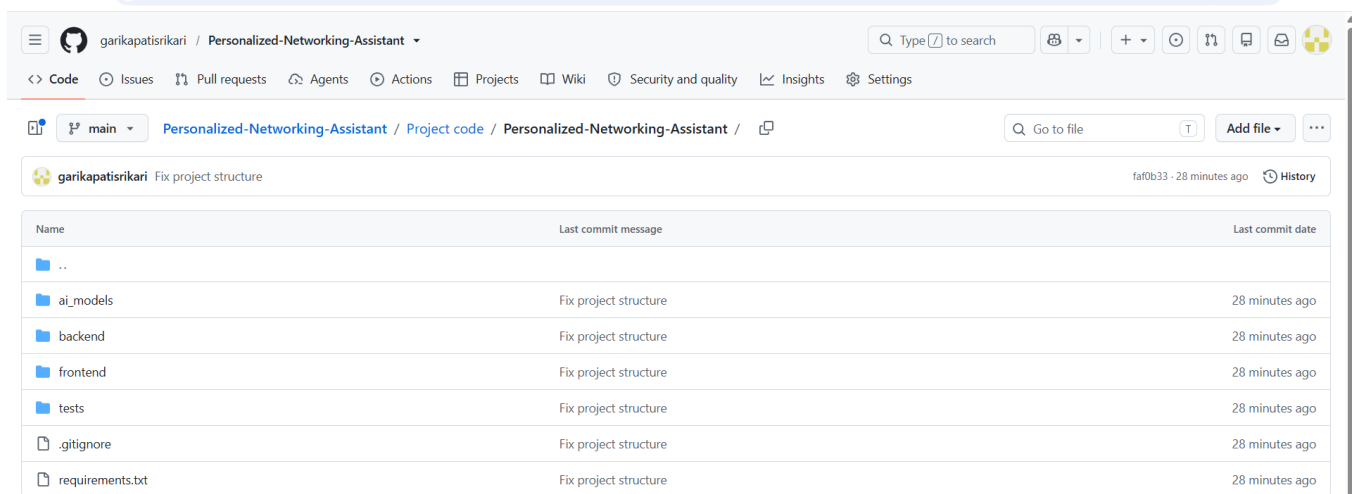
Theme Detected

Theme : Artificial Intelligence

Confidence : 0.28

Milestone 5: Testing & Deployment

The application was tested to ensure all modules worked correctly. Functional testing verified user authentication, AI conversation generation, and fact verification. The source code was uploaded to GitHub, and the application was prepared for deployment.



The screenshot shows the GitHub web interface for the repository 'Personalized-Networking-Assistant' by user 'garikapatisrikari'. The 'Code' tab is selected, displaying a file tree with folders for 'ai_models', 'backend', 'frontend', and 'tests', along with files '.gitignore' and 'requirements.txt'. All files show a commit message 'Fix project structure' from 28 minutes ago.

Name	Last commit message	Last commit date
..		
ai_models	Fix project structure	28 minutes ago
backend	Fix project structure	28 minutes ago
frontend	Fix project structure	28 minutes ago
tests	Fix project structure	28 minutes ago
.gitignore	Fix project structure	28 minutes ago
requirements.txt	Fix project structure	28 minutes ago

Conclusion

The **Personalized Networking Assistant** successfully provides AI-powered conversation generation, networking recommendations, fact verification, and conversation history management. By integrating FastAPI, Streamlit, DistilBERT, GPT-2, and the Wikipedia API, the application offers an intelligent and user-friendly solution for improving networking experiences.